

Stakeholder Engagement Plan

BRIDGING THE TOKENIZATION GAP IN AFRICAN LLMs

Project: SuperBPE for African-Language Tokenization

Team: TRI AI Cohort 10 — Team Krugger

1. Purpose of Stakeholder Engagement

The effectiveness of an African-language tokenizer cannot be determined by technical performance alone. It must also reflect how African languages are actually written, spoken, interpreted, and used by their communities.

Our stakeholder engagement approach therefore brings together native speakers, researchers, developers, educators, institutions, businesses, and language organizations to validate the technology, identify gaps, and guide future development.

Through this approach, we aim to ensure that the tokenizer is not only technically efficient but also linguistically relevant, culturally responsible, and useful in real-world applications.

2. Key Stakeholders

Stakeholder	Why They Matter	Expected Contribution
Native speakers & language communities	They understand how the languages are actually used across regions and contexts.	Validate translations, identify linguistic issues, provide feedback and contribute language data.
AI/ML researchers	They can evaluate the tokenizer scientifically and compare it with existing approaches.	Benchmarking, research validation, experimentation and technical feedback.

LLM developers	They are potential users of the tokenizer in real AI applications.	Integration testing, technical feedback and identification of practical use cases.
Universities & research institutions	They provide linguistic, technical and research expertise.	Research collaboration, datasets, validation and student/community participation.
African-language educators	They understand educational terminology and language usage in learning environments.	Validate language quality and identify educational applications.
AI startups & businesses	They can apply African-language AI to real-world products and services.	Pilot testing, use-case development and product feedback.
Language & cultural organizations	They help protect linguistic accuracy, cultural context and language diversity.	Linguistic guidance, validation and cultural review.
Open-source AI community	The project benefits from collaborative development and transparent evaluation.	Code contributions, testing, documentation and peer review.

3. Engagement Approach

Our stakeholder engagement will follow a continuous cycle:

DISCOVER - Identify the needs, challenges and expectations of different stakeholder groups.

CONSULT - Engage native speakers, researchers and domain experts to understand linguistic and technical requirements.

VALIDATE - Use native-speaker feedback and expert review to validate translations, language representations and tokenizer outputs.

TEST - Allow developers, researchers and potential users to evaluate the tokenizer against real-world use cases.

ITERATE - Use feedback and evaluation results to improve the tokenizer, datasets and evaluation methodology.

SCALE - Build partnerships that can support wider adoption, research, community participation and continued development.

4. Engagement Methods

Engagement Method	Target Stakeholders	Purpose
Community surveys	Native speakers, language communities	Understand language usage and identify pain points.
Translation validation	Native speakers, educators	Verify accuracy and naturalness of contributed translations.
Expert consultations	Researchers, linguists, educators	Obtain technical and linguistic guidance.
Tokenizer benchmarking	Researchers, developers	Compare tokenization efficiency across languages and models.
Prototype testing	Developers, businesses, researchers	Evaluate practical usability.
Community feedback sessions	Native speakers, students, educators	Gather qualitative feedback.
Open-source collaboration	Developers, researchers	Improve transparency and encourage contributions.
Partnerships	Universities, organizations, businesses	Support adoption, research and scaling.

5. Stakeholder Value Exchange

Our engagement is designed to be **two-way**.

What stakeholders gain

Native speakers - Greater representation of their languages in AI.

Researchers - New opportunities for African-language NLP research and benchmarking.

Developers - Access to an African-language-focused tokenizer and evaluation resources.

Businesses - Opportunities to explore more efficient African-language AI applications.

Universities - Opportunities for research, student participation and innovation.

Language organizations - Greater participation in the digital preservation and representation of African languages.

What the project gains

- Linguistic expertise
- Validated language data
- Real-world feedback
- Research partnerships
- Technical testing
- Adoption opportunities
- Community trust

6. Ethical Engagement Principles

Stakeholder participation will follow five principles:

Community Respect - Native speakers should have meaningful participation in how their languages are represented.

Data Responsibility - We will prioritize appropriately licensed and ethically sourced datasets.

Transparency - Data sources, preprocessing decisions and tokenizer design choices should be documented.

Inclusivity - Engagement should consider differences across languages, regions, dialects and communities.

Consent & Attribution - Where community contributions are collected, participation, consent and appropriate attribution should be clearly handled.

7. Measuring Engagement Impact

We will evaluate stakeholder engagement using measurable indicators such as:

- Number of native speakers participating.
- Number of translations submitted.
- Translation validation rate.
- Number of languages and dialects represented.
- Number of expert reviews received.
- Number of developers/researchers testing the tokenizer.
- Number of institutional or industry partnerships.
- Number of issues and improvements identified through community feedback.
- Improvements in tokenizer performance following stakeholder feedback.

8. Long-Term Engagement Vision

Our goal is to move from a **one-time project submission** to a sustainable ecosystem for African-language AI evaluation.

The people who speak African languages should have a voice in how AI learns, represents, and evaluates those languages.

SuperBPE provides the technical foundation.

EncodeLang provides the community feedback layer.

Together, they create a pathway toward more efficient, inclusive and community-informed African-language AI.